Bangladesh University of Engineering and Technology

CE 404: **Capstone Project**

Development of a Multi-storied Shopping Complex at Mohammadpur Krishi Market

Executive Summary

Submitted To:

Dr. Mehedi Ahmed Ansary Professor Dept. of Civil Engineering, BUET	Md. Amin Al Noor Assistant Professor Dept. of Civil Engineering, BUET
Snigdha Afsana Assistant Professor Dept. of Civil Engineering, BUET	Dr. Tanvir Manzur Professor Dept. of Civil Engineering, BUET

Submitted By:

1. 1904129 - Md. Atikul Islam Abhi
2. 1904122 - Md. Saif Rahman Sagor
3. 1904102 - Chinmoy Sardar
4. 1904109 - Arif Ahmed Tusar
5. 1904103 - Md. Kawsar Ahmed
6. 1904124 - Pankaj Chowdhury Partha
7. 1904110 - Rubayet Rahman Shohan

Group: 04

Section: B2

Executive Summary

1. Introduction

Established in 1978, Mohammadpur Krishi Market has been a key center for agricultural goods in Dhaka, spanning 5 acres and housing nearly 300 shops. Over the years, its functionality has been severely impacted by unplanned layouts, outdated infrastructure, poor drainage, and inadequate facilities, including a lack of parking and widespread encroachments. These issues have created challenges for vendors, shoppers, and the surrounding community.

The devastating fire on September 14, 2023, which affected 217 shops, underscored critical safety concerns, such as inadequate fire prevention measures and outdated electrical systems. Combined with flooding during heavy rains and frequent traffic congestion, these challenges highlight the urgent need for redevelopment.

Our capstone project focuses on transforming Mohammadpur Krishi Market into a modern, safe, and sustainable commercial hub. By upgrading infrastructure, enhancing fire safety, improving space utilization, and providing better amenities, we aim to create an efficient marketplace that meets the needs of vendors, customers, and the local community.

2. Project Description

1.	Name of the Project	Development of a Multi-storied Shopping Complex at Mohammadpur Krishi Market
2.	(a) Sponsoring Ministry/Division (b) Implementing Agency	Ministry of Commerce Dhaka North City Corporation (DNCC)
3.	Project Objectives (Project to be taken based on the study)	The project aims to modernize Mohammadpur Krishi Market with upgraded infrastructure, improved safety, and sustainable designs, fostering economic growth and public well-being through stakeholder collaboration.
4.	Estimated Project Cost (Taka in Crore)	144 Crore Taka
5.	Sector & Sub-Sector	Commerce

6.	Project Category (Based on Environment Conservation Rules 2023)	Red
7.	Project Geographic Location (a) Countrywide (b) Division (c) District (d) Upazila (e) Others (City Corporation/Pourashva)	 Bangladesh Dhaka Dhaka Dhaka North City Corporation
8.	Project Duration	Approximately 18 months

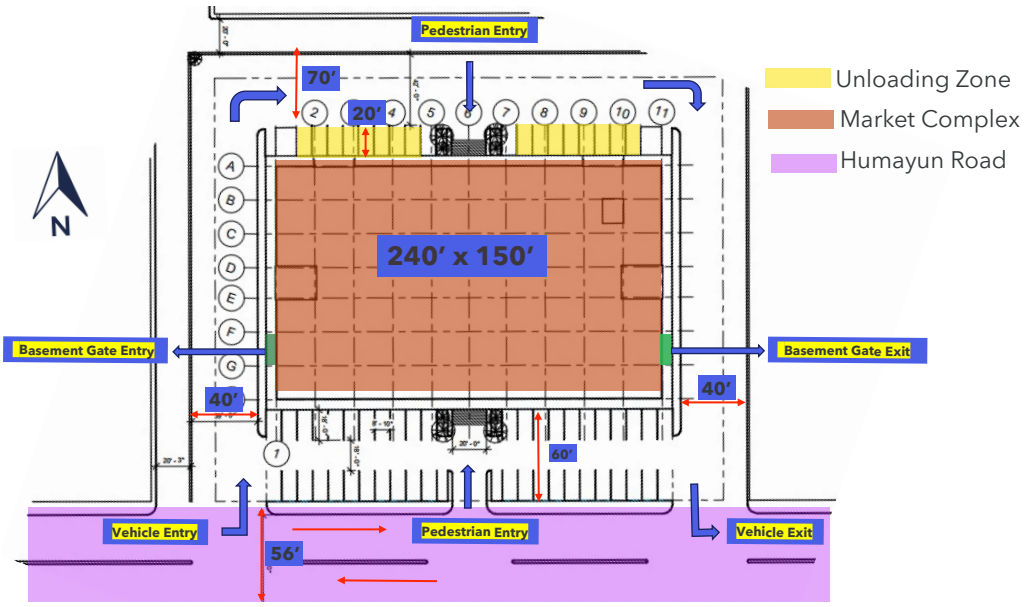


Fig.: Masterplan



Fig.: 3D View

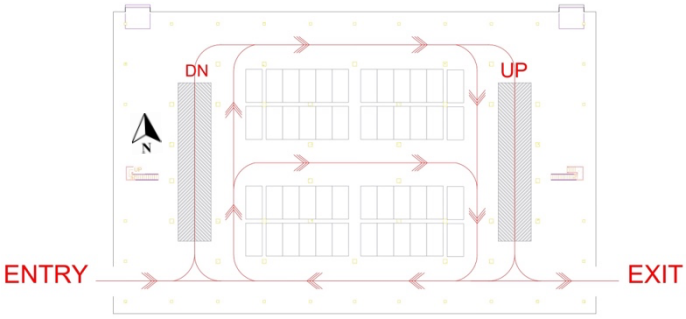


Fig.: First Basement

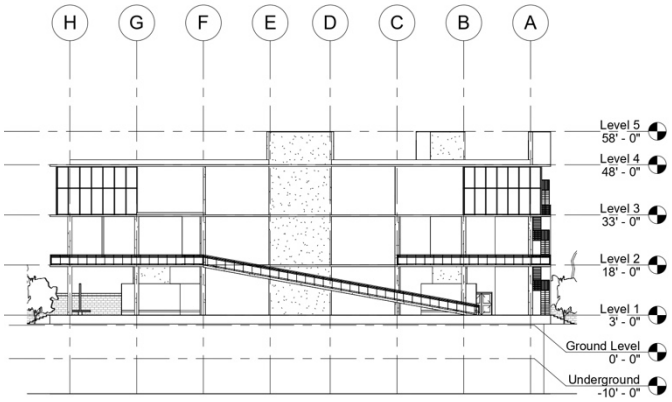


Fig.: East Elevation

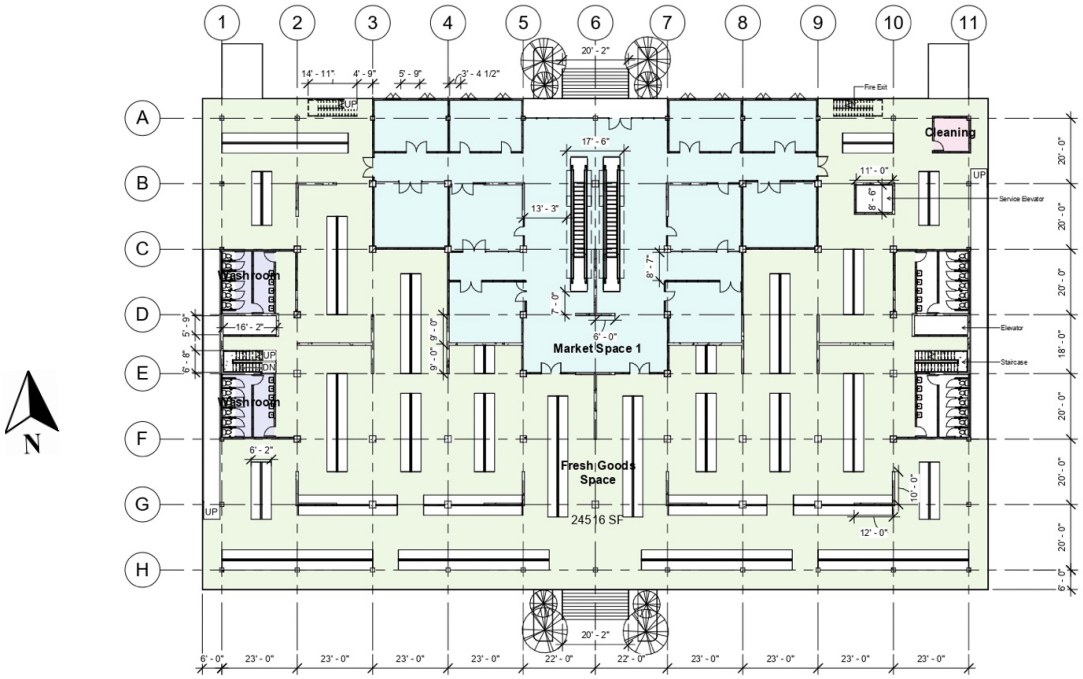


Fig.: Level 01

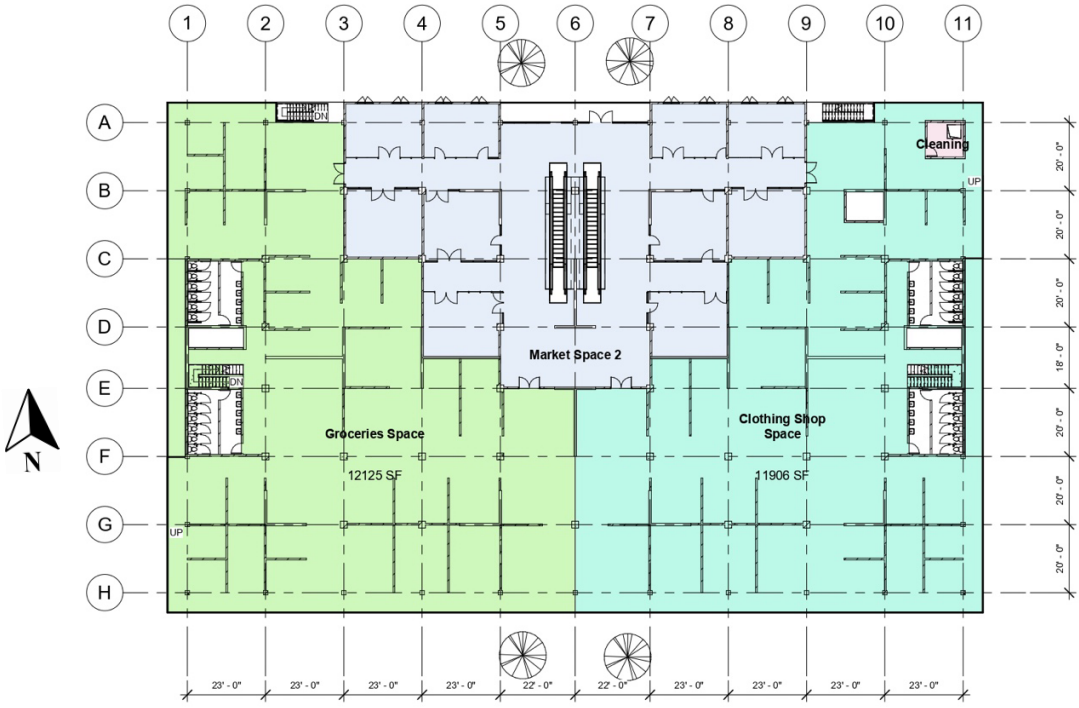


Fig.: Level 02

3. Surveys and Data

A questionnaire survey was conducted to gather user perspectives through in person. The survey included a total of **143 participants**, comprising **32 sellers** and **111 customers** of the former Krishi Market, representing a diverse range of demographic backgrounds.

Summary of Questionnaire Survey:

Survey for the Sellers:

Survey Topic	Analysis of Survey Result
Proportion of Shopkeepers Who Are Also Shop Owners	Based on the survey, 82% of the respondents indicated that they were shop owners in Krishi Market before the market was affected by the fire.
Types of Products Sold by Shopkeepers in Krishi Market	Most shopkeepers in Krishi Market sold meat and seafood (39%), followed by household cleaning products (32%) and groceries (23%). A significant portion also dealt in personal hygiene and beauty products (26%), while fewer sold fruits and vegetables (13%) or other products (10%).
Challenges Faced by Shopkeepers in Krishi Market	The most common challenge faced by shopkeepers in Krishi Market was sanitation and health code issues (68%), followed by a lack of space (55%).
Preferred Features by Users and Sellers for Market Improvement	The most desired features for market improvement were safe storage (63%) and proper sanitation (60%), reflecting the priorities of both users and sellers.
Reasons for Returning Customers to Visit Krishi Market	The key reasons for old customers to visit Krishi Market are a wide variety of products at lower prices (76%) and the availability of fresh produce (64%).

Survey for the Customers:

Survey Topic	Analysis of Survey Result
Current Shopping Locations for Customers	The majority of participants prefer shopping at supermarkets (51%), while a close percentage prefer local shops (48%).
Most Valued Feature of the Old Krishi Market According to Customers	According to customers, the best feature of the old Krishi Market was the wide variety of products available (52%), closely followed by its convenient location (47%).
Most Appreciated Feature for the New Krishi Market	The feature most appreciated for the new Krishi Market would be creating more space (60%), according to the participants.
Feature That Would Most Enhance the Shopping Experience	The feature that would most enhance the shopping experience is having enough space to move around (53%), closely followed by the introduction of home delivery service (47%).

4. Analysis of Alternatives:

Main Design: A traditional beam-supported slab resting on a mat foundation. This design is often favored for its familiarity and potentially simpler construction methods.

Alternative Design: A potentially more modern flat plate system or steel structure, supported by shallow foundation. This approach offers potential advantages in terms of aesthetics and structural efficiency but may require specialized expertise and resources.

According to various basis of comparison, the following decision matrixes have been created.

Decision Matrix : 1= least suitable 5 = most suitable

Decision matrix for choosing the type of structure for the building:

Criteria/Slab Type	Weightage (%)	Beam Supported Slab	Flat Slab	Steel structure
Cost	33	4	3	3
Construction Technology	20	4	3	2
Material Availability	7	4	4	2
Seismic Performance	14	4	2	3
Aesthetic Acceptability	7	2	3	3
Construction Time	12	3	3	4
Social Acceptability	7	4	2	2
Total	**100**	**26.18**	**20.02**	**19.46**

Beam Supported Slab = 26.18

Flat Slab = 20.02

Steel Structure = 19.46

So, Beam Supported Slab structure is selected for the project.

Decision Matrix for Choosing Foundation Type:

1= least suitable 5= most suitable

Criteria	Weightage (%)	Raft foundation	Shallow footing
Cost	33	3	4
Construction Technology	20	4	3
Material Availability	7	4	4
Seismic Performance	14	4	2

Aesthetic Acceptability	7	3	3
Construction Time	12	2	3
Social Acceptability	7	4	3
Total	**100**	**23.52**	**22.82**

Raft Foundation = 23.52

Footing = 22.82

Raft foundation is more suitable for the project.

(Pan et al. Establishing and Weighting Decision Criteria for Building System Selection in Housing Construction. Journal of Construction Engineering and Management (ASCE). 2012. 138: 1239-1250)

5. Environmental / Transportation / Social Issues

Environmental Issues:

The multi-story shopping complex at Mohammadpur Krishi Market falls under the "RED" category, necessitating an EIA to mitigate environmental impacts. Pre-construction activities may affect underground gas pipelines, while construction could obstruct drainage, cause air and noise pollution, and generate solid waste, posing risks to workers. Ecological impacts include habitat and biodiversity loss. Proper mitigation measures will address these challenges to minimize adverse effects.

Construction Related Impacts					
Environmental components	Project Activities				
	Demolition	Excavation	Drainage	Sub Structure Construction	Super Structure Construction
Air quality	✓	✓		✓	✓
Water quality			✓	✓	✓
Noise	✓	✓		✓	✓
Waste	✓	✓	✓	✓	✓

Mitigation measures:

• Use water spray to solve issues caused by dust.

• Routine equipment maintenance.

• Offer hygienic amenities.

• Reduce the needless use of sirens, horns, and alarms.

Transportation issues:

To meet the future demand of the project we will introduce-

• Road for freight vehicle

• Parking design

• Drop way for passenger.

6. Design Considerations

Material Properties	Concrete strength: $f_c' = 4$ ksi, Rebar strength: $f_y = 60$ ksi
Material Sections	Column: C1 = $18'' \times 18''$ (Corner) C2 = $24'' \times 24''$ (Edge) C3 = $30'' \times 30''$ (Interior)
	Beam: $12'' \times 24''$
	Slab: Typical slab = $6''$ 2. Stair slab = $10''$
	Foundation: Mat Foundation Thickness = 48 "
	Shore Pile: Diameter = $2'$ Length = $65'$
Assigned Loading	1. Typical Floor Live Load: 100 psf 2. Floor Finish Load: 25 psf 3. Partition Wall Load: 50 psf 4. Parking Load: 42 psf 5. Office Floor Live Load: 100 psf

Length in X Direction, L_x	240 ft
Length in Y Direction, L_y	150 ft
Total Height, h	125 ft
Story Height	15 ft (Ground, 1st and 2nd Floor) 10 ft (Basements and Other Floors)
No. of Stories	8
Exposure	A
Location	Dhaka

Wind Load Consideration:

Wind Speed	65.7 m/s
Importance Factor, I	1.15
Structure Type	Concrete Moment Resisting Frame

Earthquake Load Consideration:

Site Class	SC
Viscous Damping Ratio, ζ	5
Response Reduction Factor, R	5
Occupancy Category	III
Seismic Zone	2, Z=0.2
Lateral Seismic Force Coefficient, Sa	0.0597
Building Height Exponential, k	1.2320
Seismic Design Category	SDC C
Seismic Force Resisting System	IMRF

7. Financial and Economic Analysis

Financial Analysis:

Construction and Maintenance Cost:

Total time will be needed for completion of the project= 3 years.

Construction cost per year = 44.46 crore

Maintenance cost is taken 3% of the construction cost.

So, maintenance cost = 4 crore (for first 10 years.)

And a 15% increment is considered in the maintenance cost every 5 years.

Benefit:

There is an increment of 15% every 5 years for the next 50 years.

Here is a table with details of the several benefits-

Total Revenue		
Revenue Source	BDT (lakh)	Rate
Shop	73.5	1.5 lakh per month per 1000sft
Office	270	1.5 lakh per month per 1000sft
Food corner	44.77	3 lakh per month per 1000sft
Basement Storage	32.4	30,000 per 1000sft per month
Total Revenue (monthly)	421	
Total Revenue (yearly)	5,052	

Taking 12% discount rate

(i) Financial Net Present Value **(FNPV) = 2310.8 million BDT**

(ii) Financial Benefit Cost Ratio **(FBCR) = 2.70**

(iii) Financial Internal Rate of Return **(FIRR) = 28.33%**

Economic Analysis:

Cost Analysis

Construction cost:

To get economic cost from financial cost, we are using a standard conversion factor of 0.8.

Construction cost per year = 44.46 crore

For economic analysis,

Construction cost= 0.8* 26.68 = 35.56 crore

Operation and Maintenance cost

Maintenance cost is taken 3% of the construction cost.

So, maintenance cost = 3.2 crore (for first 5 years.)

And a 15% increment is considered in the maintenance cost every 5 years.

Benefits:

For all the benefits there is an increment of 15% every 5 years for the next 50 years.

Source	Total Revenue (BDT)
Job Opportunity	7,161,000
Tax Revenue	1,130,000
Increased property value	315,500,000
Waste and Wastewater management	1,200,000
Consumer Spending	3,650,000
Total benefit	328,641,000

Summary of economic analysis:

(i) Economic Net Present Value **(ENPV) = 1493.3 million BDT**

(ii) Economic Benefit Cost Ratio **(EBCR) = 1.08**

(iii) Economic Internal Rate of Return **(EIRR) = 18%**

8. Bill of Quantity

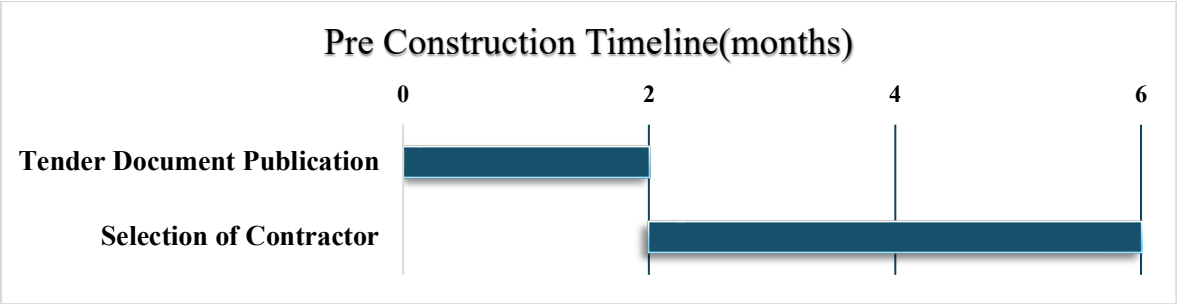
PWD (Public Works Department) grade sheet was followed during evaluation of BOQ for this project. Component wise cost and percentage of total cost breakdown are shown below

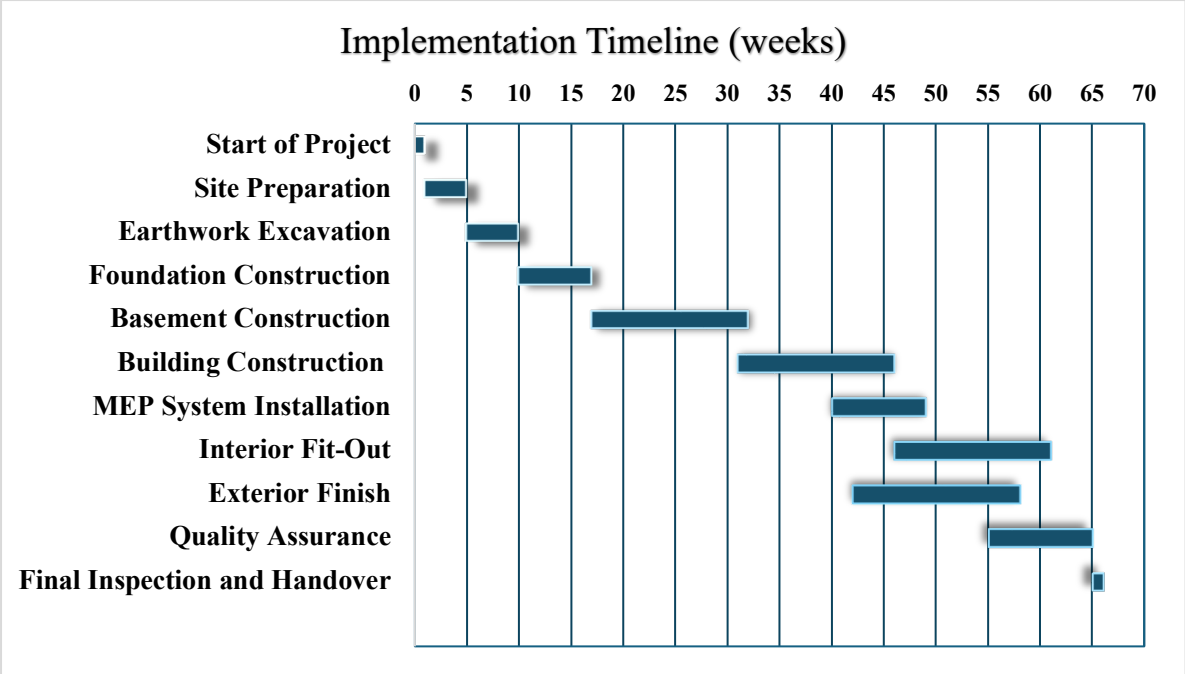
ITEMS	COST(BDT)	% of Total Cost
Shore Pile	3,790,236	0.3
Earthwork Excavation	2,139,514	0.2
Brick Flat Soling, CC	71,912,850	6.0
Tiles	63,116,886	5.3
RCC Works	322,696,622	27.1
Steel	272,976,851	22.9
Brickwork	11,059,216	0.9
Glasswork	64,628,493	5.4
Plastering	17,516,835	1.5
Wood works in Door	1,355,090	0.1
Additional Cost	357,406,207	30.0
Tests	963,960	0.1

Total Sub-Structure Cost (BDT)	**330,775,595**
Total Super-Structure Cost (BDT)	**1,104,865,619**

Grand Total (BDT)	**1,439,370,941**
Cost per sft (BDT)	**3,635**

9. Implementation Timeline





10. Conclusion

The Mohammadpur Krishi Market construction is not expected to have severe environmental consequences, with noise and dust being the primary concerns. These can be mitigated through measures to control emissions and proper maintenance of construction tools. The Environmental Impact Assessment (EIA) highlights potential issues such as air quality, traffic congestion, waste management, and community disruption, alongside strategies to address them. Environmental responsibility is a shared commitment among developers, contractors, authorities, and the community. This EIA serves as a roadmap for responsible and sustainable development, guiding the project's construction and operation.